

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0706

Jeune Landing 138 kV
October 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0706. The Interconnection Customer proposes a 278.0 MW wind facility interconnecting at the Jeune Landing 138 kV substation in the Virginia area. The request was received on November 24, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 240 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,043,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fairfax – Franconia 138 kV	Nestuca 345/138kV Autotransformer	122.7	205	32.7
Carreon – Nestuca 138 kV L	Burrows Crossing Rd – Brookridge 138 kV	118.3	190	14.9
Fairfax – Jeune Landing 138 kV line	Occoquan – Covanta Fairfax Energy 138 kV L	117.6	221	35.5
Brookridge – Franconia 138 kV Line	Nestuca 345/138kV Transformer	105.7	217	32.7
Burrows Crossing Rd – Falu 1334 Substation 138 kV	Falu 1334 – Burrows Crossing Road 138 kV Circuit 1	103.3	419	8.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fairfax – Franconia 138 kV	Rebuild line with higher-capacity conductor	\$1,970,000	100% (\$1,970,000)
Nestuca – Carreon 138 kV	Rebuild line with higher-capacity conductor	\$2,160,000	100% (\$2,160,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fairfax – Jeune Landing 138 kV	Rebuild line with higher-capacity conductor	\$1,984,000	100% (\$1,984,000)
Brookridge – Franconia 138 kV Line	Reconductor line	\$500,000	100% (\$500,000)
Burrows Crossing Rd – Falu 1334 Substation 138 kV	Reconductor line	\$696,000	62% (\$429,000)
	Total allocated to project		\$7,043,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.